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Automated analysis of surface facets: the example of cesium telluride

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High-throughput screening combined with ab initio calculations is a powerful tool to explore technologically relevant materials characterized by complex configurational spaces. Despite the impressive developments achieved in this field in the last few years, most studies still focus on bulk materials, although the relevant processes for energy conversion, production, and storage occur on surfaces. Herein, we present an automatized computational scheme that is capable of calculating surface properties in inorganic crystals from first principles in a high-throughput fashion. After introducing the method and its implementation, we showcase its applicability, focusing on four polymorphs of Cs₂Te, an established photocathode material for particle accelerators, considering slabs with low Miller indices and different terminations. This analysis gives insight into how the surface composition, accessible through the proposed high-throughput screening method, impacts the electronic properties and, ultimately, the photoemission performance. The developed scheme offers new opportunities for automated computational studies beyond bulk materials.

In the last decade, high-throughput screening methods based on density functional theory (DFT) have established themselves as a resourceful computational tool in materials science to explore complex chemical spaces^{1,2}. In addition to the ability to span a large number of configurations that can be hardly handled manually, the added value of high-throughput studies is given by automatized workflows implementing a rigorous protocol that ensures reproducible results, thus minimizing or even removing all potential issues related to the human control of calculations. The field of application of the corresponding methods encompasses the discovery of new materials for key technological applications such as optoelectronics^{3,4}, energy storage^{5–7}, photovoltaics^{8,9}, photocatalysis^{10,11}, and electron sources^{12,13}. High-throughput studies have already provided invaluable information on the structure-property relationships of several material classes, including halide perovskites^{14,15}, organic materials^{16,17}, and low-dimensional semiconductors^{18,19}.

Most of the high-throughput studies currently available for solid-state systems are focused on bulk crystals. This evidence is quite surprising, considering that most of the key processes in the fields of energy production, conversion, and storage occur on surfaces. Yet, surface calculations are significantly more demanding than their counterparts in bulk^{20,21}. This higher complexity is partly due to a large number of degrees of freedom characterizing surfaces, especially of binary and ternary compounds, such as orientation, termination, reconstruction, etc., which increases the size of the configurational space to be explored. Furthermore, numerical issues related

to the reduced dimensionality of surfaces compared to bulk crystals render corresponding self-consistent calculations harder to converge. For the aforementioned reasons, to date, very few high-throughput studies have focused on surface properties^{22,23}. Such studies include a very large number of candidate structures and supplement data analysis with machine learning algorithms, which are used to predict target observables. However, they rely on existing databases for the construction of the slabs. A general high-throughput scheme taking into account all the intricacies and pitfalls for the creation of surface facets and the characterization of their properties is, to the best of our knowledge, not yet available.

In this study, we fill this gap by presenting an automated computational procedure to calculate and post-process structural and electronic properties of surface facets from DFT in an automated fashion. We present a workflow specifically aimed to deal with slabs generated from inorganic bulk crystals and based on an in-house implemented python library¹³ interfaced with the AiiDA infrastructure^{24,25} as well as with the established libraries ase²⁶ and spglib²⁷. We describe the features of the implemented tool with the example of Cs₂Te, a known semiconductor for photocathodes in particle accelerators^{28–30}. Like most computationally predicted materials³¹, Cs₂Te is not available in single-crystalline form: it is grown via vapor deposition³⁰ and gives rise to polycrystalline samples with coexisting phases and complex surface morphologies that are hard to be systematically reproduced and characterized³². This lack of experimental information about the fundamental surface properties of Cs₂Te makes it the ideal case study for the

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computational workflow introduced in this work. The characteristics of the low-Miller-index slabs of this compound are analyzed, taking into account all possible terminations and focusing on key properties such as surface stability, band gap, and ionization potential. We relate these observables with the structural fingerprints of the slabs posing particular emphasis on the relaxation of the outermost atomic layers, which are known to most critically impact the characteristics of the surfaces. Correlations identified among computed quantities represent an added value for insight and predictions of measurable output.

Results

Workflow design

We start by presenting the high-throughput screening method specifically developed in this work to model surfaces, see Fig. 1. The implemented procedure consists of two main parts: (i) input structure generation and (ii) automatized DFT calculations and post-processing. In the following, we illustrate each step in detail.

The starting point for the high-throughput calculations is the generation of the input structures. The structural information of the parent bulk materials is mined from existing databases (e.g., Materials Project³³ and OQMD³⁴). Selected bulk crystals are initially relaxed using the same numerical parameters that are later adopted for the slabs. After this preliminary step, a symmetry analysis is performed with the Python library *spglib*²⁷ to identify the conventional unit cell of the crystal, which serves as an input to create the relevant surfaces classified by the Miller indices of the selected cutting direction. This task is performed by the *surface* function in the Python library *ase*²⁶. To take into consideration the different terminations that may occur in polyatomic materials along specific directions, a test slab is generated and segregated into atomic layers along the normal direction with respect to the surface, which is assumed to be *z*. Each of such atomic layers is considered for the slab termination. Duplicates are discarded by comparing the atomic distances within each layer as well as with respect to the atoms underneath. In a subsequent step, a symmetric, opposite termination is obtained by removing atomic layers from the bottom side of the slab until the atomic distances match with their counterparts on the other side. Each combination of Miller indices and terminations is

treated as an individual surface facet and stored as an input for the subsequent high-throughput calculations.

The procedure described above leads to the generation of data nodes (*SurfaceData* in Fig. 1) with information about the surfaces, including four different structures: (i) the repeated unit cell (including a translational vector) which represents the central part of the slab; (ii) the bottom termination of the slab; (iii) the top termination of the slab assuming a stoichiometric composition (abbreviated with *sto.* in Fig. 1); (iv) the top termination of a slab with the same terminating structure on both sides (*sym.* in Fig. 1). This way, slabs with varying thickness can be created on the fly within the workflow by appropriately combining the aforementioned building blocks.

The slabs generated according to the procedure described above represent the input for the high-throughput DFT calculations managed by the automated workflow. In the first step, the slab thickness is converged based on the method introduced in ref. 35 using the total energy of the corresponding bulk crystal as a reference. To do so, the *SurfaceData* node is used to create stoichiometric slabs which serve as an input to calculate surface energies. In the case of non-stoichiometric slabs, the bottom and top terminations compensate each other in charge to ensure that surface energy calculations are independent of the chemical potential of the constituting elements, which is chosen as a convergence criterion for the slab thickness. The thickness of the slab is then expanded until the surface energy is converged.

A slab with two identical terminations is generated, and a symmetry analysis is performed identifying the primitive slab and its layer group with *spglib*²⁷. The detected space group is mapped to the corresponding layer group, and the $\vec{k}$ -path describing the two-dimensional Brillouin zone of the surface is output (the adopted Brillouin zones are shown in the Supplementary Information, see Supplementary Fig. 1). Finally, the atomic positions are relaxed and the optimized surface slab is passed as an input to the workflow for all the subsequent calculations. The choice of dealing with charge-neutral slabs leads to uncompensated charges in the outermost atomic layers of non-stoichiometric polar surfaces, which may alter the stability and the electronic properties of the system. While several approaches have been developed to alleviate this issue^{36–39}, none of them is suitable to be embedded in automated workflows for the following reasons: they

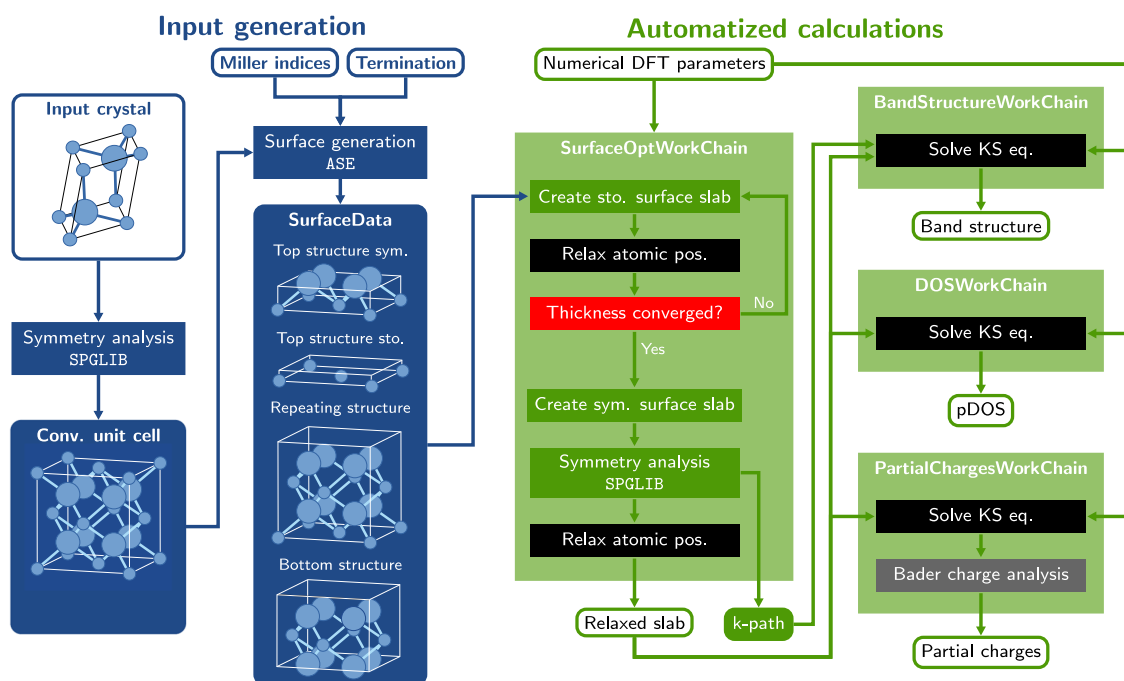


Fig. 1 | Schematic overview of the workflow. White boxes indicate input and output. Rectangular boxes denote workflow processes, and colored boxes with rounded corners show internal outputs that are used in subsequent steps of the

workflow. Black and gray boxes highlight DFT calculations and post-processing with *critic2*, respectively. Arrows highlight data provenance within the workflow.

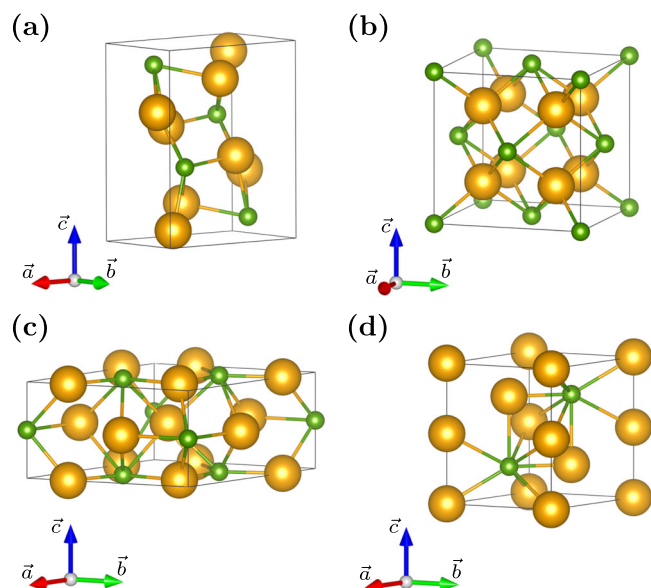


Fig. 2 | Bulk crystals. Ball-and-stick representation of the conventional cells of bulk **a** Cs_2Te with space group $Pnma$ [62], **b** Cs_2Te with space group $Fm\bar{3}m$ [225], **c** Cs_2Te with space group $P6_3/mmc$ [194], and **d** Cs_2Te with space group $R\bar{3}$ [146]. Graphics created with VESTA⁴⁴.

have been designed and tested only on specific material classes and thus cannot guarantee the generally reliable performance requested by high-throughput screening; they do not provide sufficiently high computational efficiency to be compatible with automatic algorithms. We additionally specify that surface reconstruction is not included in the current implementation: although it is crucial for an accurate description of the characteristics of the slabs, available methods to describe it require reference points that are not available a priori for any material, including Cs_2Te . Future extensions of this library will embed this and other features.

To analyze the electronic properties of the different surface facets, the band structure, the projected density of states, and the atomic partial charges are calculated. To obtain the band structure, the $\bar{k}$ -path derived in the previous step is used, while all the other parameters are the same as in ref. 13. Atomic partial charges are computed using the Bader scheme⁴⁰ as implemented in the post-processing tool `critic2`⁴¹ in combination with the Yu-Trinkle integration scheme⁴².

Cs_2Te bulk crystals

To showcase the workflow implementation, we consider cesium telluride, an established semiconducting material used as a photocathode in particle accelerators^{28–30}. We thus focus our analysis on the properties that are relevant in this specific research area, such as the stability of the surfaces and their electronic properties. To monitor the latter, we consider the frontier states, the fundamental band gap, and the ionization potential.

Experimentally, Cs_2Te has been found to crystallize in an orthorhombic lattice with space group $Pnma$ [62]⁴³. In a previous high-throughput screening study of bulk cesium telluride¹³, we identified three additional polymorphs having the same or slightly larger formation energies (see Supplementary Table 3). These polymorphs are shown in Fig. 2 and include the aforementioned orthorhombic phase, a cubic phase with space group $Fm\bar{3}m$ [225], a trigonal phase ($R\bar{3}$ [146]), and a hexagonal phase with space group $P6_3/mmc$ [194]. Given the common polycrystallinity of photocathode samples grown by vapor deposition, it is not obvious which polymorphs actually form and contribute to the surface properties. To encompass an ample range of possibilities, all four phases are taken as input in the study of surfaces.

The reference elemental phases correspond to a trigonal Te crystal with space group $P3_21$ [154]⁴⁴ and the cubic phase of Cs with space

group $Im\bar{3}m$ [229]⁴⁵. The considered bulk structures are relaxed using the same numerical parameters and workflow presented in ref. 13 except for the R2SCAN functional⁴⁶ adopted in place of the previously considered SCAN⁴⁷. The angles between the lattice parameters are fixed to preserve all crystal symmetries during relaxation. Nonetheless, we found that the trigonal phase with space group $R\bar{3}$ [146] reported in ref. 13 transforms into a hexagonal crystal with space group $P6_3/mmc$ [189], likely due to initial similarities between the two lattices. All other bulk phases maintain the original space group. Out of brevity, the bulk phases are addressed by their space group numbers in squared brackets throughout the following sections.

Surface stability and structural relaxation

Starting from the relaxed bulk structures, the computational workflow constructs slabs with low Miller indices, taking a single bulk unit cell in the periodic directions. We select the (100), (010), and (001) directions for the orthorhombic phase, (100), (110), and (111) for the cubic polymorph, as well as the (100) and (001) directions for the hexagonal systems, leading to a total of 33 different facets accounting for all possible terminations (see Supplementary Tables 4–7 in the supplementary information for the complete list). We emphasize that due to the absence of systematic experimental characterizations of Cs_2Te surfaces, we had to rely on the symmetries of the bulk phases and the general knowledge that low-Miller-index surfaces are most stable and inert⁴⁸.

We begin the analysis of the surface properties with the surface energy at 0 K, which is evaluated as the energy difference between the total energy of the slab (E^{slab}) and the chemical potentials of its elements (μ_{Cs} and μ_{Te}):

$$\gamma = \frac{1}{2A} (E^{\text{slab}} - n_{\text{Cs}}\mu_{\text{Cs}} - n_{\text{Te}}\mu_{\text{Te}}), \quad (1)$$

where n_{Cs} and n_{Te} are the number of Cs and Te atoms in the slab, respectively, and A is the surface area of the slab. Approximating the chemical potentials as the total energy per formula unit of the corresponding bulk phase,

$$2\mu_{\text{Cs}} + \mu_{\text{Te}} \approx E_{\text{Cs}_2\text{Te}}^{\text{bulk}}, \quad (2)$$

the dependency of μ_{Te} can be replaced in Eq. (1) as follows:

$$\gamma = \frac{1}{2A} (E^{\text{slab}} - n_{\text{Cs}}\mu_{\text{Cs}} - n_{\text{Te}}(E_{\text{Cs}_2\text{Te}}^{\text{bulk}} - 2\mu_{\text{Cs}})). \quad (3)$$

The condition for the surface to be stable is that the chemical potentials of the elements are limited by the chemical potential of the elemental bulk phases: $\mu_{\text{Cs}} < \mu_{\text{Cs}}^{\text{bulk}}$ and $\mu_{\text{Te}} < \mu_{\text{Te}}^{\text{bulk}}$. Approximating the chemical potentials with the total energies per atom of the elemental bulk phases ($E_{\text{Cs}}^{\text{bulk}}$ and $E_{\text{Te}}^{\text{bulk}}$) and subtracting the chemical potential of the bulk cesium phase ($\mu_{\text{Cs}}^{\text{bulk}}$), we obtain the following stability conditions for $\Delta\mu_{\text{Cs}} = \mu_{\text{Cs}} - \mu_{\text{Cs}}^{\text{bulk}}$:

$$1/2(E_{\text{Cs}_2\text{Te}}^{\text{bulk}} - E_{\text{Te}}^{\text{bulk}}) - E_{\text{Cs}}^{\text{bulk}} < \Delta\mu_{\text{Cs}} < 0.0 \quad (4)$$

which are highlighted by the two dashed, vertical lines in Fig. 4a–d.

Three different surface types can be identified (see Fig. 3 for a schematic view and Supplementary Fig. 3 for ball-and-stick representations), and their characteristics in terms of stability are visualized in Fig. 4:

- Stoichiometric surfaces (Fig. 3a): They maintain the bulk stoichiometry and are represented by horizontal lines in Fig. 4, implying that their stability does not depend on the chemical potential of their elements;
- Facets containing an excess of Cs atoms (Fig. 3b): They exhibit a negative slope in Fig. 4 since the surface energy decreases with increasing chemical potential of Cs;
- Facets containing an excess of Te atoms (Fig. 3c): They exhibit a positive slope in Fig. 4 since the surface energy increases with increasing chemical potential of Cs.

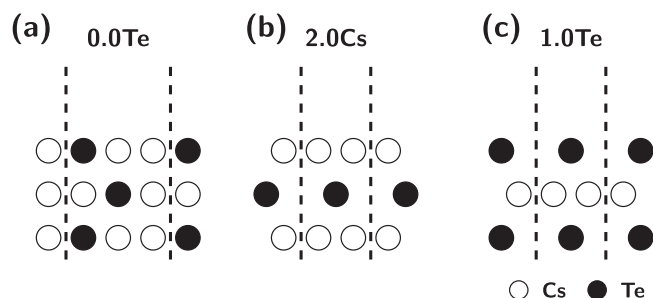


Fig. 3 | Surface types. Schematic view of **a** a stoichiometric surface, **b** a non-stoichiometric surface with excess Cs atoms, and **c** a non-stoichiometric surface with an excess of Te atoms. White and black dots indicate Cs and Te atoms, respectively, and dashed lines highlight the unit cell boundaries.

To identify the slabs according to these characteristics, we introduce the following labeling system: *excess-atom-suffix-orientation*. The *excess* index runs from 0.0 to 2.0 with steps of 0.5 and indicates the extra amount of the given *atom*, identified by its chemical symbol (Cs or Te), with respect to the bulk stoichiometry Cs_2Te . The integer *suffix* marks multiple slabs with the same amount and type of excess atoms. The *orientation* is expressed with the conventional sequence of Miller indices adopted for the given surface.

The results reported in Fig. 4 indicate the overall larger stability of the stoichiometric surface facets compared to the non-stoichiometric ones for all phases. We interpret this behavior in terms of the local charge unbalance occurring in non-stoichiometric slabs, which in turn leads to surface polarization and thus to possible energetic instabilities^{49,50}. In the orthorhombic [62] phase (Fig. 4a), this holds over almost the entire range of chemical potential of Cs ($\Delta\mu_{\text{Cs}}$), except for the extrema. The three almost degenerate stoichiometric slabs with the termination 0.0Te-1 and Miller indices (100),

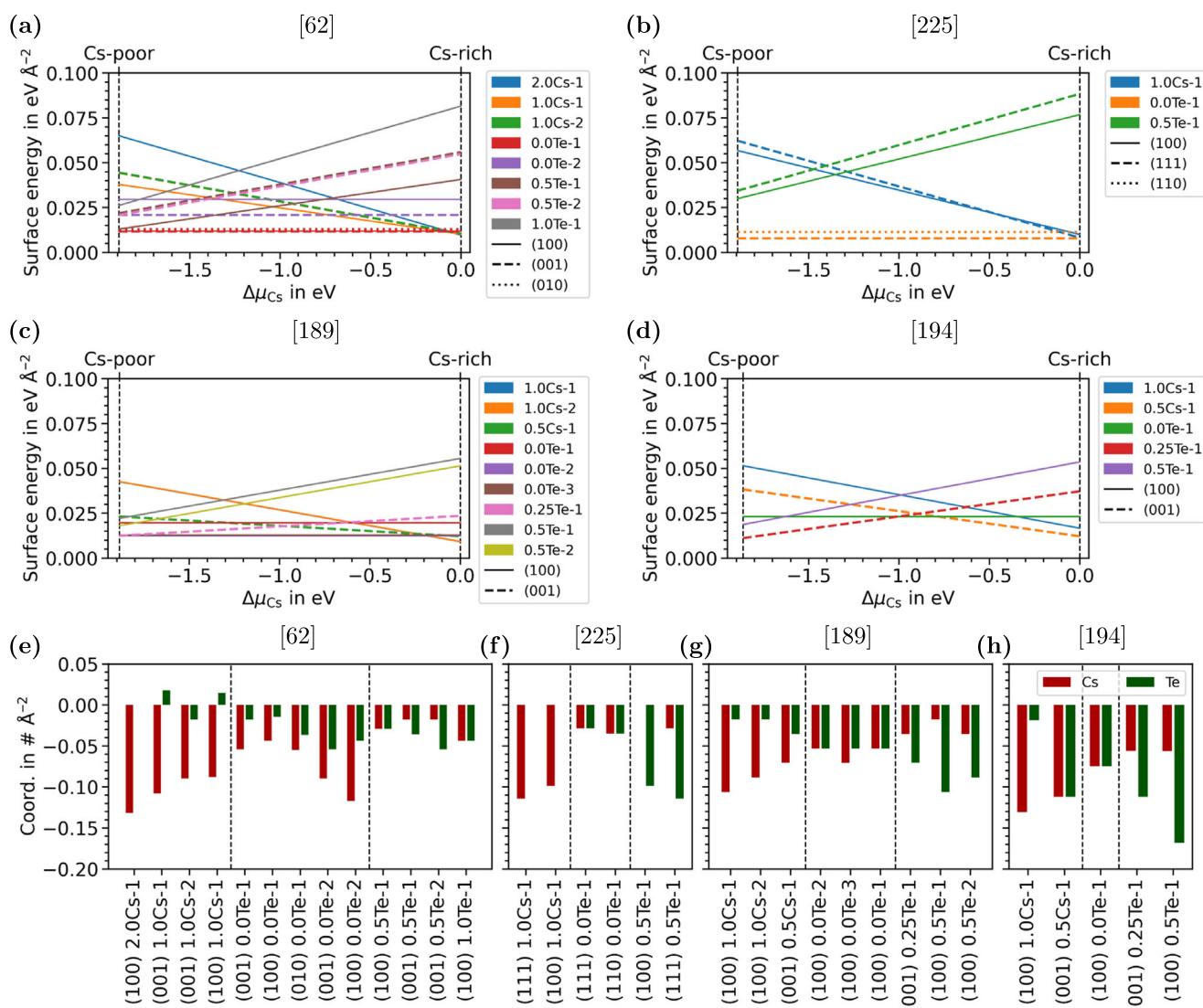


Fig. 4 | Surface stability and under-coordination. Surface energies of the considered surface facets for **a** the orthorhombic [62] phase of cesium telluride, **b** the cubic[225] phase, and the two hexagonal phases with space groups **c** $P\bar{6}2m$ [189] and **d** $P6_3/mmc$ [194]. The under-coordination per surface area as a sum over each surface site for cesium (red) and tellurium (green) atoms is referenced to the

respective sites in the bulk crystal for all surface facets of the **e** the orthorhombic phase of cesium telluride, **f** the cubic phase, and the two hexagonal phases with space groups **g** $P\bar{6}2m$ [189] and **h** $P6_3/mmc$ [194]. The vertical dashed lines in panels **e-h** mark the three different surface categories.

(010), and (001) have surface energies of 0.012–0.013 eV Å⁻² and compete against a (100) Te-rich surface, while under excess of Cs, they are overcome in energy by Cs-rich facets along (100) and (001) directions. Two additional stoichiometric surface facets are found higher in energy for the (100) and the (001) Miller indices at 0.021 eV Å⁻² and 0.030 eV Å⁻², respectively.

To rationalize the stability of different surface facets, we evaluate the sum of the coordination number over all relaxed surface sites with the neighboring atoms of the other element referenced to the respective bulk sites and normalized by the surface area of the slab, see Fig. 4e–h). Negative (positive) values correspond to an overall under-coordination (over-coordination) of the facet, highlighting that surface atoms have fewer (more) neighboring atoms than in the bulk crystal. This quantity is directly related to the number of chemical bonds broken upon the formation of the surface but also includes compensation effects due to the relaxation of the atoms. The non-stoichiometric facets are ordered based on the number of excess tellurium atoms per surface area, and the stoichiometric ones via their surface energy. To detect the surface sites in an automated fashion, the coordination and displacement of the atoms during the relaxation of the surface slab are taken as descriptors. The *econ* method⁵¹ was adopted for the determination of the coordination number with tolerance and convergence threshold set to 0.5 and 0.001, respectively⁵². If an atomic site has a different coordination in the initial slab or the atom relaxes more than 0.5 Å in the non-periodic *z*-direction, it is considered as an (under-coordinated) surface site. The coordination numbers of the four bulk phases used as reference are given in Supplementary Table 2.

Starting with the surfaces with excess cesium atoms of the orthorhombic [62] polymorph at the left of Fig. 4e), the under-coordination of cesium sites follows the same trend as the number of excess cesium atoms per surface area mounting in a maximum value of −0.13 Å⁻² for the 2.0Cs-1 facet. The under-coordination of tellurium sites, in contrast, is very small, even reaching positive values for some facets. For the stoichiometric surfaces, there is a significant downward trend of coordination towards the less stable surface facets, explaining the larger surface energies of the two facets with the 0.0Te-2 termination. As for the surfaces with excess tellurium atoms, the under-coordination is not very strongly pronounced, ranging between −0.02 and −0.04 Å⁻² for cesium and −0.03 and −0.05 Å⁻² for tellurium.

In the cubic [225] phase, the two stoichiometric slabs having surface energies of 0.008 eV Å⁻² and 0.011 eV Å⁻² are the first and second most stable ones over the entire range of $\Delta\mu_{\text{Cs}}$, except in the vicinity of $\Delta\mu_{\text{Cs}} = 0$, where two non-stoichiometric surfaces cut along the (100) and (111) directions with an excess of Cs atoms compete against them (see Fig. 4b). The Te-rich slabs considered for this system are the least stable under Cs-rich conditions while they are more favorable than those with an excess of Cs up to values of $\Delta\mu_{\text{Cs}} \approx -1.5$. The low surface energy of stoichiometric facets is also emphasized by their low under-coordination (Fig. 4f) in contrast to the non-stoichiometric facets of the polymorph.

Moving on to the hexagonal phases, we notice that for the one belonging to the space group *P62m* [189], two stoichiometric slabs with the same surface energy of 0.013 eV Å⁻² are more stable than the non-stoichiometric ones except for values of $\Delta\mu_{\text{Cs}}$ very close to 0.0 (see Fig. 4c). In this scenario, one of the (100)-slabs with an excess of one Cs atom reaches the lowest value of surface energy. Te-rich facets are again the least stable ones except for the most Cs-poor region with $\Delta\mu_{\text{Cs}} < -1.5$. Turning to the under-coordination of the different slabs represented in Fig. 4g), the three facets with excess cesium atoms exhibit a strong under-coordination of cesium like the slabs of the orthorhombic and cubic phase. All stoichiometric surfaces show similar under-coordination values: the less stable (100) 0.0-Te1 facet cannot even be distinguished from the more stable (100) 0.0-Te2 and (100) 0.0-Te3 facets. A possible reason for this behavior is given by the subtle structural differences between the slabs, e.g., different bond lengths that are not captured by the coordination number but that nonetheless affect the stability. The surfaces with excess tellurium atoms are characterized by an under-coordination of these sites, which is less pronounced for the more stable 0.25Te-1 termination.

Finally, among the surfaces cut from the hexagonal [194] phase, we cannot identify the most stable one over a broad range of $\Delta\mu_{\text{Cs}}$ (Fig. 4d). In this case, the only stoichiometric slab has the lowest surface energy merely for $\Delta\mu_{\text{Cs}} \approx -1.0$ with an absolute value of 0.023 eV Å⁻². Comparing this facet with the most stable stoichiometric facets of the previous polymorphs, the surface energy is significantly larger. The more pronounced under-coordination of this slab in comparison to the stoichiometric slabs of the other polymorphs, which exhibit lower surface energies, can explain this behavior. At the extrema of the plot, the most favorable surfaces are those cut along the (001) direction with a 0.25 excess of Te for $\Delta\mu_{\text{Cs}} < -1.0$ and with a 0.5 excess of Cs for $\Delta\mu_{\text{Cs}} > -1.2$. The slabs obtained along the (100) direction are clearly less stable than the other ones. Additional details about the coordination of the surface atoms and their rearrangement upon optimization are provided in Supplementary Fig. 4.

We conclude this section with a few considerations about surface bond formation emerging from the analysis of Fig. 4e–h). The undercoordination of Cs and Te under the conditions discussed above leads to dangling bonds at the surface, which are expected to be saturated by atmospheric pollutants in ambient conditions or by residual atoms in ultrahigh vacuum. The chosen simulation cell consisting of a single replica of the bulk unit cell in the periodic directions of the slab does not offer room for additional patterns also related to surface reconstruction. These aspects go beyond the scope of this work and will be addressed in a future analysis.

Electronic properties of selected surface facets

In the next step, we analyze the electronic properties of representatives of the three different surface types of the orthorhombic [62] phase cut in the (100) direction (see Fig. 3). While the 2.0Cs-1 and the 0.0Te-1 facets are the most stable ones with $\Delta\mu_{\text{Cs}}$ up to −0.12 eV and −0.04 eV, respectively, the 0.5Te-1 facet is slightly higher in energy than the 0.0Te-1 one under Te-rich conditions. Band structures and projected density of states (PDOS) of slabs computed for other structures are reported in the Supplementary information (Supplementary Figs. 5 and 6). Inheriting their main features from the bulk material^{13,53}, in all the considered band structures, the highest valence states show little dispersion and are dominated by the *p*-orbitals of the Te atoms. The lowest conduction states, on the other hand, have a pronounced parabolic shape at the Γ -point, stemming from the *s*-orbitals of the Cs atoms. Overall, this results in a direct band gap of 2.27 eV at the Γ -point and in an indirect, fundamental band gap of 2.17 eV between *X* and Γ for the orthorhombic and cubic bulk phases, respectively (the band-structure plots are in Supplementary Fig. 2).

Differences among the electronic structures of the surfaces arise from the different coordination of the surface atoms, as shown in the PDOS, where the contributions of Cs and Te atoms on the outermost surface layers (surface) and in the inner part of the slab (bulk) are shown (see Fig. 5). The surface facets with an excess of Cs atoms are represented by the (100) 2.0Cs-1 surface of the orthorhombic polymorph (Fig. 5a). Here, the partially occupied *s*-orbitals of the Cs atoms at the surface dominate the region around the Fermi energy. Their influence is so strong that the systems are metallic. The ionization potential (*IP*) can be readily accessed via the energy difference between the vacuum level (E_{vac}) given by the system's electrostatic potential (red dashed line) and the energy of the valence band maximum (E_{VBM})

$$IP = E_{\text{vac}} - E_{\text{VBM}} \quad (5)$$

with a value of 1.36 eV for the (100) 2.0Cs-1 facet.

Looking at the band structure of the stoichiometric slab (Fig. 5b), we notice that in the surfaces with stoichiometric composition, the *p*-orbitals of the surface Te atoms play a role in shifting the valence band maximum (VBM) upwards, resulting in a band gap of 2.00 eV which is smaller than in the bulk crystal. The ionization potential of the surface facet increases in comparison to the metal-terminated facet and amounts to 2.59 eV. Along the (010) and (001) directions, the slabs with 0.0Te-1 termination have band gaps of 2.00 and 2.14 eV and *IP* of 3.09 and 3.03 eV, respectively. These

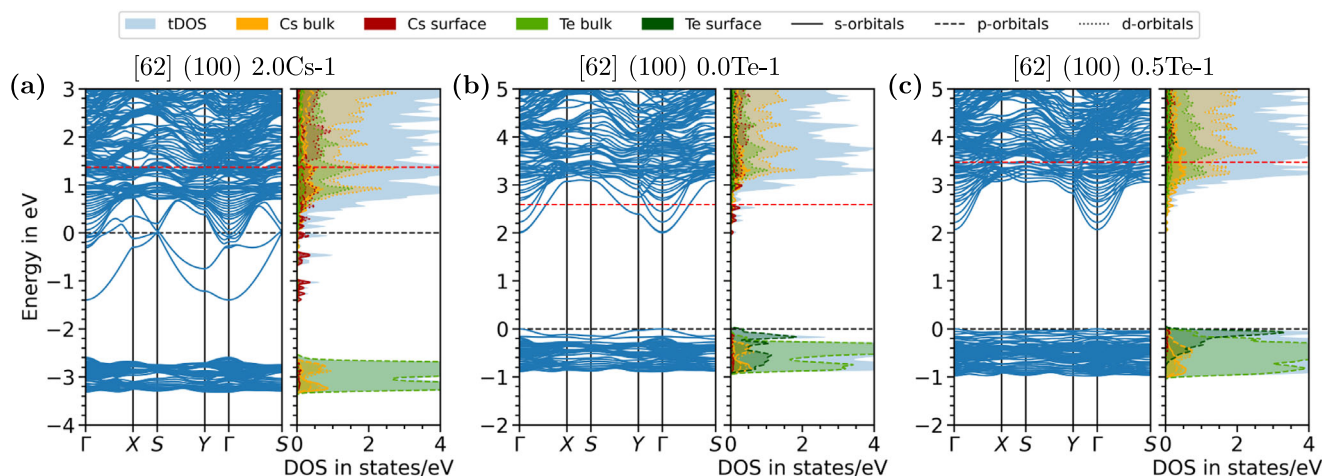


Fig. 5 | Electronic band structures and PDOS of selected facets. Electronic band structures and projected density of states (PDOS) of three facets with Miller indices (100) of the orthorhombic polymorph representing different surface types: **a** 2.0Cs-1 having an excess of Cs atoms, **b** stoichiometric 0.0Te-1, and **c** 0.5Te-1 facet with an

excess of Te atoms. The VBM (or the Fermi energy for the metallic system in **a**) is set to 0 eV and marked by a black dashed line; the vacuum level is highlighted by a red dashed line. The PDOS shows the contributions of the two elements (Cs and Te), distinguishing between “bulk” and “surface” sites.

values slightly differ from the work functions calculated in ref.⁵⁴ for Cs₂Te surfaces using the PBE functional⁵⁵. A possible reason for this discrepancy can be the adopted approximation for the exchange-correlation potential.

Turning now to the surface facet with an excess of Te, the contributions from the Te surface atoms are dominant in the highest valence region (see Fig. 5c). In these systems, the lowest unoccupied states actually overlap with the highest occupied ones, resulting in partially filled valence bands. This issue has already been discussed for non-stoichiometric surface facets of CsK₂Sb⁵⁶: At room temperature, these states are subject to the smearing of the Fermi surface, and thus, this electronic instability is expected to be resolved. With this background knowledge, we deliberately evaluate the band gap from the VBM even though it is predicted to be unoccupied from DFT at 0 K. The gap, evaluated from the unoccupied VBM, turns out to be 2.07 eV with an ionization potential of 3.47 eV.

Experimental reference values to compare with are scarce. Velazquez et al. measured an initial work function of 2.3 eV and 2.8 eV in aged Cs₂Te samples using Kelvin probe microscopy⁵⁷. The lower value of the work function (2.3 eV) is in between the values of *IP* calculated here for the surface facets with excess Cs atoms and the stoichiometric surface facets; the higher value (2.8 eV) is within the range obtained for the stoichiometric surface facets. The polycrystalline character of the material and the possibility that several surface facets may locally form and contribute to the measured values should be considered in this comparison.

Correlation between electronic properties and surface terminations

In the final step of our analysis, we examine the electronic properties of cesium telluride surface facets that are relevant to their photoemission performance. In particular, we aim to correlate the characteristics of the surfaces and their termination with the *IP* and the fundamental band gap evaluated as the energy difference between the conduction band minimum (CBm) and the VBM. We consider the number of excess Te atoms normalized by the surface area as a universal descriptor that can be calculated for the three different surface types, see Fig. 6. This quantity is defined as the slope of the surface energy with respect to the chemical potential visualized in Fig. 4.

As mentioned before, in the case of excess Te atoms (negative excess of Cs atoms), the band-gap values are zero due to the overlap between the lowest unoccupied and the highest occupied ones, which results in partially filled valence bands. Here, we consider the gap slightly above the VBM for comparison. Comparing the results obtained for the band gaps with those harvested for the *IP*, we notice, in general, a similar trend (see Fig. 6). Expectedly, the metallic slabs with an excess of Cs atoms exhibit the smallest

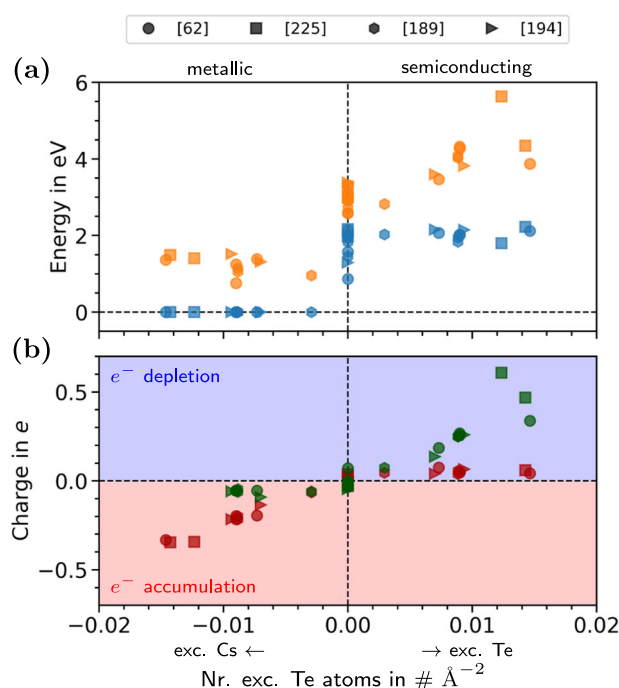


Fig. 6 | Electronic properties of all facets. **a** Band gap (blue) and ionization potential (orange) of the considered slabs as a function of the number of excess Te atoms at the surface; **b** average charge accumulation of surface Cs (red) and Te atoms (green) at the surface sites with respect to the number of excess Te atoms per surface area.

values of *IP*, on the order of 0.5–1.5 eV, in agreement with the intuition that metallic systems should be prone to emit electrons with limited energetic costs. Following the same line of reasoning, the large-band-gap facets with an excess of Te atoms are characterized by sizeable *IP* values ranging from about 3 eV up to above 5 eV. Clearly, these systems do not represent favorable candidates for photocathodes, although, considering their relatively large surface energies (Fig. 4), it is quite unlikely that such systems will form. Finally, for the stoichiometric slabs, we find intermediate *IP* values ranging from about 2.5–3.5 eV. Given the overall lowest values of surface energies associated with these systems, we expect actual cesium telluride samples to exhibit *IP* in this range.

To add another piece to this analysis, we inspect the average charge accumulation on the atoms occupying the surface sites of each system. In Fig. 6b), we report the average charge accumulation on Cs and Te atoms estimated as the difference between the Bader partial charge⁴⁰ on each species at the surface sites and in the corresponding bulk crystal. The trend obtained for the Cs atoms indicates negative values of the partial charges for all the slabs characterized by an excess of Cs atoms. This behavior can be ascribed to the lack of neighboring Te atoms balancing the charge distribution through bond formation, resulting in electron accumulation at the surface. For the stoichiometric slabs, the excess charges on Cs atoms approach zero. Upon excess of Te atoms at the surface, the extra charges on Cs are positive but very close to zero (see Fig. 6b). Turning now to the Te excess charges, we notice a similar trend as the one seen in Fig. 6a) for the *IP*. For slabs with an excess of Cs atoms, the Te atoms are slightly negatively charged, exhibiting values around $-0.1 e$ or lower. In the stoichiometric surfaces, the excess charges on Te are close to zero, while they undergo a dramatic increase upon increasing the excess of Te atoms.

The similarity between Fig. 6a, b suggests a correlation between electron accumulation or depletion and the energy requested by the system to remove an electron, which is likely to be also the case for related materials. While some surface facets included in this study turn out to be unstable, the range of band gaps and ionization potentials achieved by different outermost terminating atomic layers may give useful insights on how to tweak the electronic properties by either adjusting the experimental conditions or actively functionalizing the surface. In this context, 2d-material functionalizations to reduce the work functions^{58,59} and single crystalline samples⁶⁰ with better control on the formed surface termination are relevant developments to improve the material's performance as a photocathode.

Discussion

To sum up, we have introduced a high-throughput method to calculate the surface properties of crystalline materials aimed at unraveling the complex relationships between the composition and the stability as well as the electronic properties of the various facets that can be formed from a bulk. The presented implementation is embedded in a recently developed library for automatized workflow and post-processing tools for DFT calculations¹³ based on the infrastructure *AiiDA* as well as on *ase* and *spglib*. The applicability of this tool is demonstrated with the exemplary case of cesium telluride, an established semiconducting material employed as a photocathode in particle accelerators. Specifically, we focused on four polymorphs of Cs_2Te identified as particularly stable in a previous high-throughput study on the bulk phases¹³.

We found that the stoichiometric surface facets are the most stable ones over a large range of chemical potential values for the orthorhombic, cubic, and one of the two examined hexagonal polymorphs. The surface facets terminating with an excess of Te atoms become stable only toward Te-rich conditions. Under Cs-rich conditions, instead, surface facets with an excess of Cs atoms turn out to be slightly more stable than the others for most of the considered phases. The band gaps and the ionization potentials of the surfaces, subsequently considered in this analysis, strongly correlate with the number of excess atoms at the surface. Facets with additional Cs atoms exhibit a metallic character with a diminishing band gap and a reduced *IP* resulting from the partially filled *s*-orbitals of the alkali species. For the stoichiometric surface facets, on the other hand, a broad range of band-gap and *IP* values spanning more than 1 eV are obtained. The origin of this behavior is the under-coordination of the surface sites for the different facets, which in turn give rise to electronic levels in the middle of the gap, i.e., gap states. Finally, surface facets with an excess of Te atoms approach the band-gap values of the corresponding bulk crystal and show the largest *IP* among the considered systems. In this case, an approximately linear dependence between the excess atoms per surface area and the ionization potential is noticeable.

In conclusion, with the implementation of an automatized workflow specifically designed for surfaces, we offer a new tool for the analysis of systems and properties that are particularly relevant in the photoemission of

photocathodes as well as in many other key processes for energy production, conversion, and storage^{61,62}. Additional features that are planned to be incorporated in the near future include an enhanced description of surface properties, taking into account, e.g., asymmetric slabs, surface states, and surface reconstruction^{63–65}, as well as the integration with machine learning algorithms for the analysis of the output properties. With the available functionalities and these perspectives in reach, the presented infrastructure enriches the library of tools for automated material discovery and characterization, bridging the incoming era of artificial intelligence.

Methods

Computational details

The DFT calculations^{66,67} embedded in the presented workflow are performed using the CP2K code⁶⁸, implementing Gaussian and plane wave basis-set schemes⁶⁹. The TZV2P-MOLOPT basis set was chosen for both elements (Cs and Te) with a cutoff and a relative cutoff energy of 550 Ry and 100 Ry, respectively. Core electrons are represented by dual-space Goedecker-Teter-Hutter pseudopotentials⁷⁰ specifically parameterized for the SCAN functional⁴⁷. With background knowledge of the good performance of SCAN for this class of materials⁵³, we adopted R2SCAN to overcome the numerical instabilities featured by SCAN when dealing with low-dimensional systems, including slabs^{46,71}. We checked the agreement between the results delivered by these two functionals for the four Cs_2Te bulk crystals considered in this work (see Supplementary Table 1 and Fig. 2).

In the surface calculations, the slab is converged starting from a minimal thickness of 20 Å and monitoring the surface energy with a threshold of $10^{-4} \text{ eV Å}^{-2}$. Atomic positions within the slabs are relaxed until forces are lower than 0.01 eV Å^{-1} . To include a sufficient amount of vacuum between neighboring facets, the lattice parameter in the normal direction is set to 2.1 times the slab thickness. The Martyna–Tuckerman solver⁷² is applied to treat the direction normal to the surface as non-periodic. With this procedure, neighboring facets are completely decoupled, and the amount of vacuum separating them no longer plays a role in the calculations. The reciprocal space in the two periodic directions is sampled using a Monkhorst–Pack mesh⁷³ with the *k*-points separated from each other by 0.15 Å^{-1} , while only the Γ -point is sampled along the non-periodic direction. The density of the *k*-grid used in the surface calculations is the same as in the corresponding bulk samples. The projected density of states is obtained by simulating the system in a supercell with a lattice parameter larger than 25 Å in the periodic directions and with the corresponding Brillouin zone sampled at Γ only.

Data availability

The datasets generated and analyzed during the current study are available in [Zenodo](#).

Code availability

The code adopted to perform the first-principles calculations presented in this work, CP2K⁶⁸, is publicly available open-source at <https://www.cp2k.org/> (visited on January 24, 2024). Access to the library for automatized calculations can be provided to qualified researchers upon reasonable request to the corresponding authors.

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Author contributions

H.D.S. designed the work, developed and implemented the computational tools, produced the data, analyzed and curated them, and wrote the first version of the paper, including the preparation of all figures. C.C. supervised the project, harvested funding, discussed the results, and finalized the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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